

Chapter 6, Problem 14P

Problem

Series-connected 20- and 60-pF capacitors are placed in parallel with series-connected 30- and 70-pF capacitors. Determine the equivalent capacitance.

Step-by-step solution

Step 1 of 4

Consider the following data:

Series connected 20 pF and 60 pF capacitors are placed in parallel with series-connected 30 pF and 70 pF capacitors.

Draw the capacitors series parallel connected circuit diagram using given data:

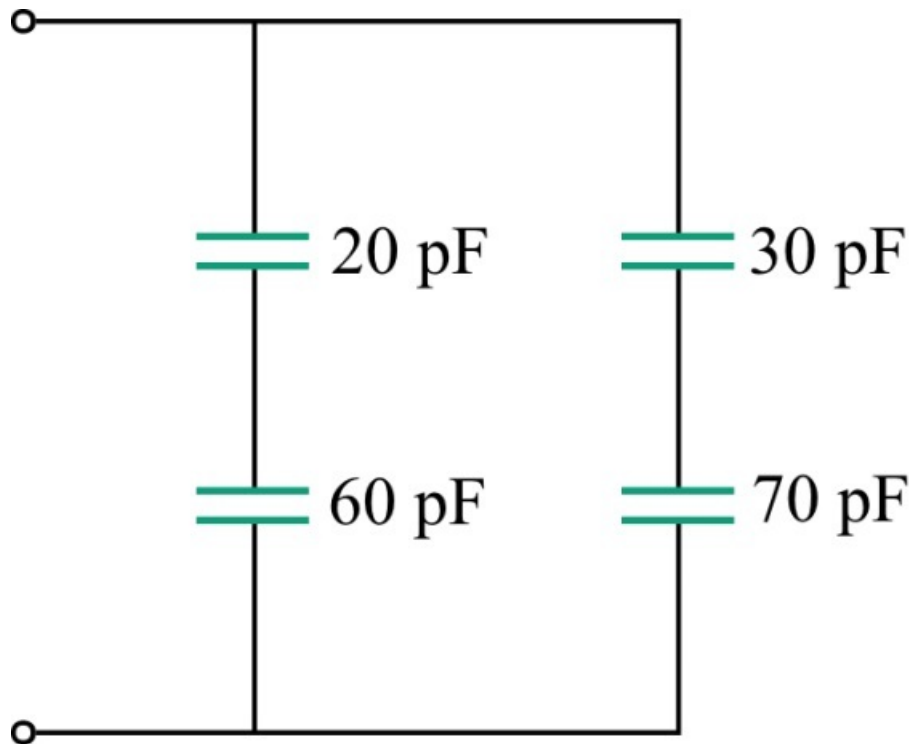


Figure 1

Step 2 of 4

In the Figure 1, 20 pF and 60 pF capacitors are in series, calculate the equivalent capacitance and assigned as C_{eq1} .

$$\begin{aligned}C_{eq1} &= \frac{(20 \text{ pF})(60 \text{ pF})}{20 \text{ pF} + 60 \text{ pF}} \\&= \frac{1200 \text{ pF}}{80} \\&= 15 \text{ pF}\end{aligned}$$

Similarly, 30 pF and 70 pF capacitors are in series, calculate the equivalent capacitance and assigned as C_{eq2} .

$$\begin{aligned}C_{eq2} &= \frac{(30 \text{ pF})(70 \text{ pF})}{30 \text{ pF} + 70 \text{ pF}} \\&= \frac{2100 \text{ pF}}{100} \\&= 21 \text{ pF}\end{aligned}$$

Step 3 of 4

Draw the modified circuit diagram after converting series capacitors in to equivalent capacitance.

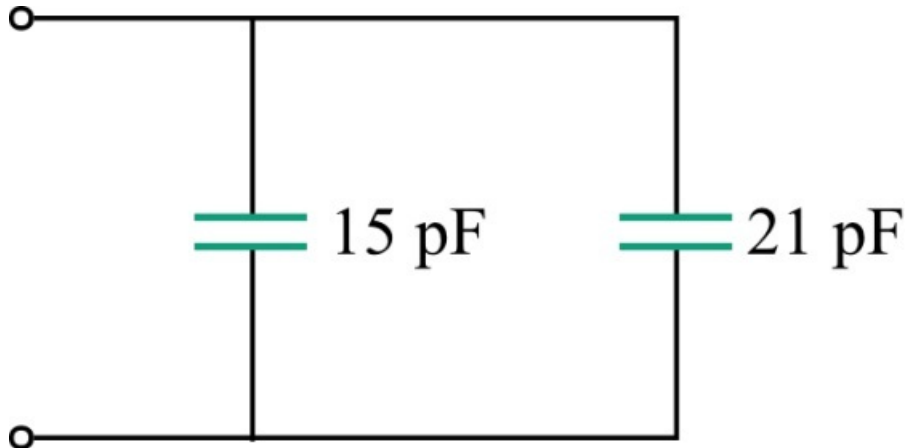


Figure 2

Step 4 of 4

In the Figure 2, 15 pF and 21 pF capacitors are in parallel and calculate the equivalent capacitance.

$$\begin{aligned}C_{eq} &= 15 \text{ pF} + 21 \text{ pF} \\&= 36 \text{ pF}\end{aligned}$$

Therefore, the equivalent capacitance, C_{eq} is 36 pF.